5 Discrete transforms

5.1 The $\mathcal{Z}$-transform

Integral transforms allow us to simplify problems involving functions of a continuous variable t, but in many practical situations we don't have access to a function defined for every value of t. Instead, we may only have knowledge of a function at certain time intervals and we have to adapt our techniques to deal with this discrete data.

The discrete analogue of functions are *sequences*, which we denote by $a = (a_n)_{n=0}^{\infty}$. If we wish to think of these as signals, we can think of the signal updating at even time intervals, with the value of the signal at a time $n\Delta t$ being a_n. Some simple examples are

$$(n)_{n=0}^{\infty} = (0,1,2,3,\ldots), \quad (\delta_n)_{n=0}^{\infty} = (1,0,0,0,\ldots), \quad (\lambda^n)_{n=0}^{\infty} = (1,\lambda,\lambda^2,\lambda^3,\ldots), \tag{5.1}$$

where $\lambda \in \mathbb{C}$ is some constant. Given two sequences $a = (a_n)_{n=0}^{\infty}$ and $b = (b_n)_{n=0}^{\infty}$, we can define linear combinations of them component-wise

$$\alpha a + \beta b = (\alpha a_n + \beta b_n)_{n=0}^{\infty} = (\alpha a_0 + \beta b_0, \alpha a_1 + \beta b_1, \ldots), \tag{5.2}$$

making the space of sequences indexed by $\mathbb{Z}_{\geq 0}$ a vector space.

Definition 5.1. Let $a = (a_n)_{n=0}^{\infty}$ be a sequence. The *$\mathcal{Z}$-transform* of a is given by

$$A(z) = \mathcal{Z}[a] = \sum_{n=0}^{\infty} a_n z^{-n}, \tag{5.3}$$

where z is a complex variable.

Much like the Laplace transform, the $\mathcal{Z}$-transform only really makes sense for values of z such that the summation converges. This defines the *region of convergence* of a given $\mathcal{Z}$-transform. In particular, if a sequence $(a_n)_{n=0}^{\infty}$ is bounded by

$$|a_n| \leq Ar^n, \quad A, r \geq 0, \tag{5.4}$$

making it an *exponential type* sequence, we have

$$|A(z)| \leq \sum_{n=0}^{\infty} |a_n||z^{-n}| \leq A \sum_{n=0}^{\infty} \left(\frac{r}{|z|}\right)^n. \tag{5.5}$$

If $r/|z| < 1$, the right-hand side converges and we can use the formula for geometric sums to get

$$|A(z)| \leq \frac{A}{1 - r/|z|} = \frac{A|z|}{|z| - r}. \tag{5.6}$$

Therefore, the region of convergence is given by $|z| > r$. In other examples, the region of convergence can often be inferred by using a known Maclaurin series or checking where the poles of the summed series lie.

Example 5.1. (i) For the sequence $(\lambda^n)_{n=0}^{\infty}$, the $\mathcal{Z}$-transform is given by

$$\mathcal{Z}[\lambda^n] = \sum_{n=0}^{\infty} \lambda^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{\lambda}{z}\right)^n. \tag{5.7}$$

Noting that the binomial expansion $1/(1-q) = 1+q+q^2+\ldots$ is valid for $|q|<1$, we can rewrite the above series as

$$\mathcal{Z}[\lambda^n] = \frac{1}{1-\lambda/z} = \frac{z}{z-\lambda}, \tag{5.8}$$

as long as $|z| > |\lambda|$.

(ii) Using the above, we can derive the $\mathcal{Z}$-transforms for $\cos \alpha n$ and $\sin \alpha n$. Concretely, we have

$$\mathcal{Z}[\cos \alpha n] = \frac{1}{2}(\mathcal{Z}[\mathrm{e}^{\mathrm{i}\alpha n}] + \mathcal{Z}[\mathrm{e}^{-\mathrm{i}\alpha n}]) = \frac{1}{2}\left(\frac{z}{z-\mathrm{e}^{\mathrm{i}\alpha}} + \frac{z}{z-\mathrm{e}^{-\mathrm{i}\alpha}}\right). \tag{5.9}$$

Combining the fractions, we can simplify this as

$$\mathcal{Z}[\cos \alpha n] = \frac{z^2 - \frac{1}{2}z(\mathrm{e}^{\mathrm{i}\alpha} + \mathrm{e}^{-\mathrm{i}\alpha})}{z^2 - z(\mathrm{e}^{\mathrm{i}\alpha} + \mathrm{e}^{-\mathrm{i}\alpha}) + 1} = \frac{z^2 - z\cos\alpha}{z^2 - 2z\cos\alpha + 1}. \tag{5.10}$$

Similarly, the $\mathcal{Z}$-transform of $\sin \alpha n$ is

$$\mathcal{Z}[\sin \alpha n] = \frac{z\sin\alpha}{z^2 - 2z\cos\alpha + 1}. \tag{5.11}$$

The poles of these functions occur at $z_{\pm} = \mathrm{e}^{\pm \mathrm{i}\alpha}$, so the region of convergence is given by $|z| > 1$.

5.2 Relation to Laplace transform ★

The $\mathcal{Z}$-transform can be thought of as a discrete version of the Laplace transform. To see this, we first define the *Dirac comb* $\text{Ш}_T(t)$[2] to be

$$\text{Ш}_T(t) = \sum_{n=0}^{\infty} \delta(t - nT), \tag{5.12}$$

where $T > 0$ is a constant. Then for any given function $g(t)$, we can calculate the Laplace transform of $g(t)\text{Ш}_T(t)$ as

$$\mathcal{L}[g(t)\text{Ш}_T(t)] = \sum_{n=0}^{\infty}\left(\int_{0^-}^{\infty} \delta(t-nT)g(t)\mathrm{e}^{-st}\,\mathrm{d}t\right) = \sum_{n=0}^{\infty} g(nT)\mathrm{e}^{-snT}. \tag{5.13}$$

[2]This is the Cyrillic letter 'sha' Ш, chosen purely because it looks a bit like a comb.

sequence	$\mathcal{Z}$-transform
a_n	$A(z)$
$\alpha a_n + \beta b_n$	$\alpha A(z) + \beta B(z)$
$\lambda^n a_n$	$A\left(\frac{z}{\lambda}\right)$
na_n	$-zA'(z)$
$a'_n = \begin{cases} a_{n-k} & n \geq k, \\ 0 & n < k. \end{cases}$	$z^{-k}A(z)$
a_{n+1}	$zA(z) - a_0 z$
a_{n+k}	$z^k A(z) - \sum_{j=0}^{k-1} a_j z^{k-j}$
$(a * b)_n$	$A(z)B(z)$

Table 3: Transformation rules for $\mathcal{Z}$-transforms. Here α, β are arbitrary complex constants, $\lambda \neq 0$ and $k \geq 1$.

Integrating against the Dirac comb has the effect of picking out the value of $g(t)\mathrm{e}^{-st}$ at evenly spaced intervals $t = nT$ for $n \in \mathbb{Z}_{\geq 0}$. If we now let $z \doteq \mathrm{e}^{sT}$, this becomes

$$\mathcal{L}[g(t)\mathrm{III}_T(t)] = \sum_{n=0}^{\infty} g(nT)z^{-n}, \tag{5.14}$$

matching the $\mathcal{Z}$-transform of the sequence $a_n = g(nT)$. While we won't use this fact, the relation between z and s provides a way of translating stability results from Laplace transforms to $\mathcal{Z}$-transforms, among other things. If you're interested in this, consider how the region of stability for continuous LTI systems $\{s : \mathrm{Re}(s) < 0\}$ changes under the map $s \mapsto \mathrm{e}^{sT}$ and compare with the results later in this chapter.

5.3 Transformation properties

Yes, this is the third section called this. Yes, you know the drill.

As with the other transforms we've seen thus far, the $\mathcal{Z}$-transform is linear, meaning that

$$\mathcal{Z}[\alpha a + \beta b] = \alpha A(z) + \beta B(z), \tag{5.15}$$

for any complex numbers α and β. Instead of scaling a sequence by multiplying by a constant, we can also consider scaling each term of a sequence $a = (a_n)_{n=0}^{\infty}$ exponentially as $b = (\lambda^n a_n)_{n=0}^{\infty}$. Then the $\mathcal{Z}$-transform of b is given by

$$\mathcal{Z}[\lambda^n a_n] = \sum_{n=0}^{\infty} a_n \lambda^n z^{-n} = \sum_{n=0}^{\infty} a_n \left(\frac{z}{\lambda}\right)^{-n} = A\left(\frac{z}{\lambda}\right). \tag{5.16}$$

Similarly, we could scale the sequence linearly by defining $c = (na_n)_{n=0}^{\infty}$. In this case, the $\mathcal{Z}$-transform of c is related to that of a by

$$\mathcal{Z}[na_n] = \sum_{n=0}^{\infty} na_n z^{-n} = z \sum_{n=0}^{\infty} a_n (nz^{-n-1}) = -z \frac{\mathrm{d}}{\mathrm{d}z}\left(\sum_{n=0}^{\infty} a_n z^{-n}\right) = -zA'(z). \tag{5.17}$$

Here we've recognised nz^{-n-1} as the derivative of $-z^{-n}$ and written the series in such a way to use this fact. This provides the analogue of the Laplace transform identity $\mathcal{L}[tf(t)] = -F'(s)$, only here we also have to introduce another factor of z to match up with the definition of the $\mathcal{Z}$-transform.

We can also consider shifting (or translating) the sequence, but since our sequence is indexed by $\mathbb{Z}_{\geq 0}$, we need to distinguish in which direction we're shifting. First consider delaying the sequence by k-many steps, so that the new sequence is

$$b_n = \begin{cases} a_{n-k} & n \geq k, \\ 0 & n < k. \end{cases} \tag{5.18}$$

Here we've been careful to include k-many zeroes, since a_n is not defined for $n < 0$. Since we are essentially moving the whole sequence k-steps to the right, this is often called a ***right translation***. We will often abuse notation and write $(a_{n-k})_{n=0}^{\infty}$ to denote the right translation of $(a_n)_{n=0}^{\infty}$ by k-many steps with the understanding that we are including k-many zeroes at the start of the sequence. The $\mathcal{Z}$-transform of a right translation is given by

$$\mathcal{Z}[b_n] = \sum_{n=k}^{\infty} a_{n-k} z^{-n} = z^{-k} \sum_{n=k}^{\infty} a_{n-k} z^{-(n-k)}. \tag{5.19}$$

By introducing the variable $m = n - k$, we can rewrite the final sum to get

$$\mathcal{Z}[b_n] = z^{-k} \sum_{m=0}^{\infty} a_m z^{-m} = z^{-k} A(z). \tag{5.20}$$

Therefore, a right translation of k-many steps just introduces an extra factor of z^{-k} to the $\mathcal{Z}$-transform. On the other hand, a ***left translation*** $b_n = (a_{n+k})_{n=0}^{\infty}$ is a little more complicated. In particular, the $\mathcal{Z}$-transform is given by

$$\mathcal{Z}[a_{n+k}] = \sum_{n=0}^{\infty} a_{n+k} z^{-n} = z^k \sum_{n=0}^{\infty} a_{n+k} z^{-(n+k)} = z^k \sum_{m=k}^{\infty} a_m z^{-m}. \tag{5.21}$$

Now we see the issue: we are missing terms including a_n for $n < k$, so we need to add them by hand in order to write this in terms of $A(z)$. This is easily done

$$\mathcal{Z}[a_{n+k}] = z^k \left(\sum_{m=0}^{\infty} a_m z^{-m} - \sum_{m=0}^{k-1} a_m z^{-m} \right) = z^k A(z) - \sum_{m=0}^{k-1} a_m z^{-m+k}. \tag{5.22}$$

For concreteness, let's take $k = 2$, then this rule reads as

$$\mathcal{Z}[a_{n+2}] = z^2 A(z) - z^2 a_0 - z a_1. \tag{5.23}$$

This should remind you of the Laplace transform, where derivatives get mapped to multiplication by s^m up to some boundary terms. For the $\mathcal{Z}$-transform, the role of these boundary terms is played by a_n for $n < k$.

Finally, we also have a discrete version of the convolution theorem.

Theorem 5.1. Given two sequences $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$, the $\mathcal{Z}$-transform of the discrete convolution

$$(a * b)_k = \sum_{n=0}^{k} a_n b_{k-n}, \tag{5.24}$$

is given by

$$\mathcal{Z}[(a * b)_k] = A(z)B(z). \tag{5.25}$$

Proof. Substituting the series expressions for $A(z)$ and $B(z)$ in $A(z)B(z)$ gives

$$A(z)B(z) = \left(\sum_{n=0}^{\infty} a_n z^{-n}\right)\left(\sum_{m=0}^{\infty} b_m z^{-m}\right) = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} a_n b_m z^{-n-m}. \tag{5.26}$$

By introducing the variable $k = m + n$, we can rewrite the sum as

$$A(z)B(z) = \sum_{k=0}^{\infty}\left(\sum_{n=0}^{k} a_n b_{k-n}\right) z^{-k}. \tag{5.27}$$

We recognise the term in the bracket as $(a * b)_k$, so we identify this with the $\mathcal{Z}$-transform of $(a * b)$. □

Example 5.2. (i) We will use the transformation properties above to derive the $\mathcal{Z}$-transform of the sequence

$$a_n = \begin{cases} 1/n & n \geq 1, \\ 0 & n = 0. \end{cases} \tag{5.28}$$

First notice that the sequence $(na_n) = (0, 1, 1, \ldots)$ is the right-translation of the constant sequence $u = (1, 1, 1, \ldots)$. Therefore,

$$-zA'(z) = \mathcal{Z}[na_n] = z^{-1}\mathcal{Z}[u_n] = \frac{1}{z-1}. \tag{5.29}$$

Rearranging for $A'(z)$, we get

$$A'(z) = -\frac{1}{z(z-1)} = \frac{1}{z} - \frac{1}{z-1,} \tag{5.30}$$

which, upon integration, gives

$$A(z) = C + \log z - \log(z-1) = C + \log\left(\frac{z}{z-1}\right). \tag{5.31}$$

In order to determine the constant of integration C, we need to use the fact that $a_0 = 0$ somehow. We can do so by noticing that $A(z) \to a_0$ as $z \to \infty$. Since $\log(z/(z-1)) \to 0$ as

$z \to \infty$, we have that $C = 0$. All in all, we have

$$\mathcal{Z}[a_n] = \log\left(\frac{z}{z-1}\right) = -\log(1 - z^{-1}). \tag{5.32}$$

(ii) Consider the constant sequence $a = (1)_{n=0}^{\infty}$ and $b = (n)_{n=0}^{\infty}$. The convolution of these sequences is

$$(b * a)_n = \sum_{k=0}^{n} k = \tfrac{1}{2}n(n+1). \tag{5.33}$$

The convolution theorem therefore states that

$$\mathcal{Z}\left[\tfrac{1}{2}(n^2 + n)\right] = \frac{z}{z-1} \cdot \frac{z}{(z-1)^2} = \frac{z^2}{(z-1)^3}. \tag{5.34}$$

Now using the linearity of the $\mathcal{Z}$-transform we can rearrange for $\mathcal{Z}[n^2]$ to get

$$\mathcal{Z}[n^2] = \frac{2z^2}{(z-1)^3} - \mathcal{Z}[n] = \frac{2z^2}{(z-1)^3} - \frac{z}{(z-1)^2} = \frac{z^2 + z}{(z-1)^3}. \tag{5.35}$$

As an exercise, you can check that this matches the result you'd obtain by using the identity $\mathcal{Z}[n^2] = -zB'(z)$.

5.4 Difference equations

The discrete counterpart to a differential equation is a *difference equation*. These relate the value a_n of a sequence $(a_n)_{n=0}^{\infty}$ to other terms in the sequence. For example, the Fibonacci sequence $(f_n)_{n=0}^{\infty}$ is defined by the difference equation

$$f_{n+2} = f_{n+1} + f_n, \tag{5.36}$$

subject to the conditions $f_0 = 0$ and $f_1 = 1$. More generally, difference equations are useful in modelling systems whose inputs and outputs update in discrete jumps, such as in digital signal processing or population growth models. In the latter, the discrete jumps can represent different generations of a species, with the value of a sequence a_n modelling the population of said species in the nth generation.

So how do we solve difference equations? As you've probably guessed, we can use the $\mathcal{Z}$-transform together with the rules for left and right translations to turn this into a much simpler problem. Taking the Fibonacci example, we have

$$\mathcal{Z}[f_{n+2}] = \mathcal{Z}[f_{n+1}] + \mathcal{Z}[f_n] \implies z^2F(z) - z = zF(z) + F(z). \tag{5.37}$$

Rearranging for $F(z)$ then gives

$$F(z) = \frac{z}{z^2 - z - 1} \tag{5.38}$$

Now we need to invert this to find the general formula for f_n. Just like the continuous case, partial fractions is the key here, but there's a nice trick that we can use to make things a little easier. Recall that we know the $\mathcal{Z}$-transforms

$$\mathcal{Z}[\lambda^n] = \frac{z}{z-\lambda}, \quad \mathcal{Z}[n\lambda^n] = \frac{\lambda z}{(z-\lambda)^2}, \tag{5.39}$$

and so on. The fact that z appears in the numerator of all of these transforms is giving us the hint that we want to write the partial fractions decomposition with a factor of z out front. Therefore, we write the partial fractions decomposition of $F(z)$ as

$$F(z) = z\left(\frac{A}{z-\alpha} + \frac{B}{z-\beta}\right), \tag{5.40}$$

with α and β the distinct roots of $z^2 - z - 1 = 0$. Then comparing this with the $\mathcal{Z}$-transforms above, we have the general solution

$$f_n = A\alpha^n + B\beta^n, \tag{5.41}$$

where A and B are the coefficients obtained from the partial fractions procedure. As an exercise, you can find A, B, α and β and show that

$$f_n = \frac{\varphi^n - \tilde{\varphi}^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \quad \tilde{\varphi} = \frac{1-\sqrt{5}}{2}. \tag{5.42}$$

This is known as ***Binet's formula*** and it relates the Fibonacci numbers to the golden ratio φ and its conjugate $\tilde{\varphi} = -1/\varphi$.

Example 5.3. Let $(a_n)_{n=0}^{\infty}$ be a sequence that satisfies the following difference equation

$$a_{n+2} + 3a_{n+1} + 2a_n = 0, \tag{5.43}$$

with initial conditions $a_0 = a_1 = 1$. Taking the $\mathcal{Z}$-transform gives

$$z^2A(z) - z^2 - z + 3(zA(z) - z) + 2A(z) = 0, \tag{5.44}$$

which we rearrange to get

$$A(z)(z^2 + 3z + 2) = z^2 + 4z \Longrightarrow A(z) = z\left(\frac{z+4}{z^2+3z+2}\right). \tag{5.45}$$

We can factorise the denominator as $(z+1)(z+2)$ and perform partial fractions on the bracketed part as

$$\frac{z+4}{(z+1)(z+2)} = \frac{A}{z+1} + \frac{B}{z+2} \Longrightarrow z + 4 = A(z+2) + B(z+1). \tag{5.46}$$

Taking $z = -1$ causes the term including B to vanish, while taking $z = -2$ causes the A term to vanish. Therefore,

$$(z = -1): \quad A = 3 \qquad\qquad (z = -2): \quad B = -2. \tag{5.47}$$

As such, the partial fraction decomposition of $A(z)$ is

$$A(z) = \frac{3z}{z+1} - \frac{2z}{z+2}. \tag{5.48}$$

Comparing with the $\mathcal{Z}$-transforms we derived earlier (or on the formula sheet), we obtain

$$a_n = 3(-1)^n - 2(-2)^n. \tag{5.49}$$

We can check the initial conditions as a sanity check

$$a_0 = 3 \cdot 1 - 2 \cdot 1 = 1, \qquad a_1 = 3(-1) - 2(-2) = 1, \tag{5.50}$$

as expected.

Example 5.4. We can use the $\mathcal{Z}$-transform to solve systems of difference equations. Let $a = (a_n)_{n=0}^{\infty}$ and $b = (b_n)_{n=0}^{\infty}$ be two sequences satisfying the difference equations

$$\begin{aligned} a_{n+1} &= 2a_n - b_n, \\ b_{n+1} &= -a_n + b_n, \end{aligned} \tag{5.51}$$

with $a_0 = 0$ and $b_0 = 1$. Taking the $\mathcal{Z}$-transform of both equations gives a set of linear equations for $A(z)$ and $B(z)$

$$\begin{aligned} zA(z) &= 2A(z) - B(z), \\ zB(z) - z &= -A(z) + B(z). \end{aligned} \tag{5.52}$$

Rearranging the top equation for $A(z)$ and substituting this into the bottom equation gives

$$A(z) = -\frac{B(z)}{z-2} \implies B(z)(z-1) = z + \frac{B(z)}{z-2}. \tag{5.53}$$

Now we can rearrange for $B(z)$ to get

$$B(z) = z\left(\frac{z-2}{z^2-3z+1}\right) = z\left(\frac{C}{z-\alpha} + \frac{D}{z-\alpha^{-1}}\right), \tag{5.54}$$

where $\alpha = \frac{1}{2}(3+\sqrt{5})$ is the larger root of $z^2 - 3z + 1 = 0$. As a somewhat annoying exercise, you can show that

$$C = \frac{5-\sqrt{5}}{10}, \qquad D = \frac{5+\sqrt{5}}{10}, \tag{5.55}$$

giving

$$B(z) = \frac{5-\sqrt{5}}{10}\frac{z}{z-\alpha} + \frac{5+\sqrt{5}}{10}\frac{z}{z-\alpha^{-1}}. \tag{5.56}$$

n	0	1	2	3	4	5	6	7	8
a_n	0	−1	−3	−8	−21	−55	−144	−377	−987
b_n	1	1	2	5	13	34	89	233	610

Table 4: The solutions to the system of difference equations in eq. (5.51) for $0 \leq n \leq 8$.

Therefore, the sequence b is given by

$$b_n = \frac{5-\sqrt{5}}{10}\alpha^n + \frac{5+\sqrt{5}}{10}\alpha^{-n}. \tag{5.57}$$

We're not done though, we need to find a_n as well. We know that $A(z) = -B(z)/(z-2)$, so we have

$$A(z) = -z\left(\frac{1}{z^2-3z+1}\right) = \frac{1}{\sqrt{5}}\frac{z}{z-\alpha^{-1}} - \frac{1}{\sqrt{5}}\frac{z}{z-\alpha}. \tag{5.58}$$

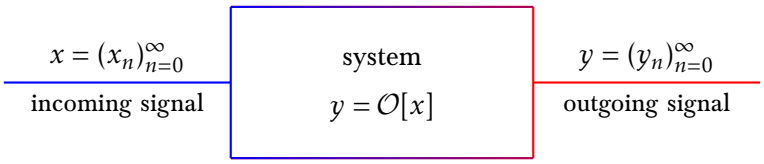
Therefore, the sequence a is given by

$$a_n = \frac{1}{\sqrt{5}}\alpha^{-n} - \frac{1}{\sqrt{5}}\alpha^n. \tag{5.59}$$

This all looks a bit complicated, but calculating the first few values of a_n and b_n displays a nice pattern, as shown in table 4. (Compare these with the first 16 Fibonacci numbers!)

5.5 Return of LTI systems

We now consider the discrete version of LTI systems, which we visualise as a black box as before, with an incoming signal x_n being mapped to the outgoing signal y_n through an operation $\mathcal{O} : x_n \mapsto y_n$.



The requirement that a system is LTI and causal is completely analogous to the continuous case:

- *Linearity.* For any two incoming signals x_n and x'_n with corresponding outgoing signals y_n and y'_n and any choice of constants $\alpha, \beta \in \mathbb{C}$, $\mathcal{O}$ satisfies

$$\mathcal{O}[\alpha x_n + \beta x'_n] = \alpha\mathcal{O}[x_n] + \beta\mathcal{O}[x'_n] = \alpha y_n + \beta y'_n. \tag{5.60}$$

- *Time invariance.* Delaying an incoming signal by k-many steps delays the outgoing signal by k-many steps with no other changes. That is,

$$\mathcal{O}[x_{n-k}] = y_{n-k}, \tag{5.61}$$

  for any incoming signal x_n.

- *Causality.* If x and x' are two incoming signals such that $x_k = x'_k$ for $0 \leq k \leq n$, then the corresponding outgoing signals satisfy $y_n = y'_n$.

The last point is slightly different to how we phrased causality in the continuous case, but it is equivalent. Essentially, since the two incoming signals match up to a certain point, the system cannot distinguish them as it only knows about past values of the incoming signal and knows nothing about how they differ in the future.

Theorem 5.2. The output of any causal discrete LTI system is given by the convolution

$$(x * g)_n = \sum_{k=0}^{n} x_k g_{n-k}, \tag{5.62}$$

where x_n is the incoming signal and g_n is the impulse response $\mathcal{O}[\delta_n]$.

Proof. Let $x = (x_0, x_1, x_2, \ldots)$ be the incoming signal of a causal LTI system and y the corresponding output. We define a new signal $x' = (x_0, x_1, \ldots, x_n, 0, 0, \ldots)$ which matches x up to the nth step. We can decompose x' into a finite sum of impulses

$$x' = x_0(\delta_m)_{m=0}^{\infty} + x_1(\delta_{m-1})_{m=0}^{\infty} + \ldots + x_n(\delta_{m-n})_{m=0}^{\infty} = \sum_{k=0}^{n} x_k(\delta_{m-k})_{m=0}^{\infty}. \tag{5.63}$$

The nth step of the corresponding output is therefore

$$y'_n = \sum_{k=0}^{n} x_k \mathcal{O}[(\delta_{n-k})_{n=0}^{\infty}] = \sum_{k=0}^{n} x_k g_{n-k} = (x * g)_n, \tag{5.64}$$

where we've used the linearity property in the first equality and the time invariance property to establish the last equality. By causality, this must equal y_n. Since we can do this procedure for any value of n, this gives y_n for every $n \in \mathbb{Z}_{\geq 0}$. □

We now restrict to the class of LTI systems whose outgoing signals are given by the solution to a difference equation

$$a_N y_{n+N} + a_{N-1} y_{n+N-1} + \ldots + a_1 y_{n+1} + a_0 y_n = x_n, \tag{5.65}$$

subject to the conditions

$$y_n = 0, \quad \forall n < N. \tag{5.66}$$

These boundary conditions are required by linearity and state that the outgoing signal is completely at rest until N-many time steps have passed. The nice thing about this assumption is that the $\mathcal{Z}$-transform of eq. (5.65) is particularly simple

$$Y(z)(a_N z^N + a_{N-1} z^{N-1} + \ldots + a_1 z + a_0) = X(z), \tag{5.67}$$

leading to the solution

$$Y(z) = \frac{X(z)}{a_N z^N + a_{N-1} z^{N-1} + \ldots + a_1 z + a_0}. \tag{5.68}$$

When $x = (\delta_n)_{n=0}^{\infty}$, $X(z) = 1$ and we identify the *transfer function* $G(z)$ with the rational function

$$G(z) = \frac{1}{a_N z^N + a_{N-1} z^{N-1} + \ldots + a_1 z + a_0}, \tag{5.69}$$

so $Y(z) = G(z)X(z)$, as expected.

Example 5.5. Consider the LTI system governed by the difference equation

$$y_{n+2} - 4y_{n+1} + 4y_n = x_n, \tag{5.70}$$

subject to $y_0 = y_1 = 0$. Taking the $\mathcal{Z}$-transform, we get

$$Y(z)(z^2 - 4z + 4) = X(z) \Longrightarrow Y(z) = \frac{X(z)}{(z-2)^2}. \tag{5.71}$$

Taking $X(z) = 1$ identifies the transfer function with $1/(z-2)^2$. This almost looks like the $\mathcal{Z}$-transform of the sequence $b = (n2^{n-1})_{n=0}^{\infty}$, but we're missing the factor of z in the numerator. By looking at the formula sheet, we see that this corresponds to an additional right translation by 1 step, so the inverse sequence is

$$g_n = \begin{cases} (n-1)2^{n-2} & n \geq 1, \\ 0, & n = 0. \end{cases} \tag{5.72}$$

The first few terms of the sequence are therefore $g = (0, 0, 1, 4, 12, \ldots)$. Therefore, the full solution for y_n is given by

$$y_n = \sum_{k=0}^{n} x_k g_{n-k} = \sum_{k=0}^{n-1} x_k (n-k-1) 2^{n-k-2}. \tag{5.73}$$

Here the upper limit is $n-1$, since $g_0 = 0$. For example, if $x_k = 2^k$, the corresponding solution is

$$y_n = \sum_{k=0}^{n-1} (n-k-1) 2^{n-2} = 2^{n-2} \sum_{k=0}^{n-1} (n-k-1). \tag{5.74}$$

It can be shown that the final sum equates to $n(n-1)/2$, giving the solution $y_n = 2^{n-3}n(n-1)$. As a check, we can substitute this into the difference equation with $n > 0$ to get

$$\begin{aligned} y_{n+2} - 4y_{n+1} + 4y_n &= 2^{n-1}(n+2)(n+1) - 2^n(n+1)n + 2^{n-1}n(n-1), \\ &= 2^{n-1}(n^2 + 3n + 2 - 2(n^2+n) + n^2 - n), \\ &= 2^{n-1} \cdot 2, \\ &= 2^n, \end{aligned} \tag{5.75}$$

verifying the solution.

5.6 Stability analysis

Now we turn to the stability of discrete LTI systems, the definition of which is much the same as before, just adapted to sequences.

Definition 5.2. (i) A discrete LTI system is *stable* if

$$\lim_{n\to\infty} g_n = 0, \tag{5.76}$$

where g is the impulse response.

(ii) The system is ***bounded-input, bounded-output (BIBO)*** stable if any bounded incoming signal results in a bounded outgoing signal. That is, for any incoming x_n satisfying $|x_n| \leq A$ for all n, there exists $B \geq 0$ such that $|y_n| \leq B$ for all n.

One can reformulate BIBO stability as

$$\|g\|_1 = \sum_{n=0}^{\infty} |g_n| < \infty. \tag{5.77}$$

This is equivalent to our definition since, assuming that $|x_n| \leq A$ for some constant A, we have that the outgoing signal satisfies

$$|y_n| \leq \sum_{k=0}^{n} |x_k||g_{n-k}| \leq A \sum_{k=0}^{n} |g_{n-k}| = A \sum_{m=0}^{n} |g_m| \leq A\|g\|_1, \tag{5.78}$$

where we've defined $m = n-k$ in the middle equality. Demanding that $\|g\|_1$ is finite then implies that y_n is bounded, showing that $\|g\|_1 < \infty$ implies BIBO stability as defined above. Similarly, it is easy to show that assuming bounded inputs and outputs demands $\|g\|_1 < \infty$, showing the equivalence of the two definitions.

Again, for LTI systems with rational transfer functions these are equivalent notions, but for non-rational transfer functions they are not. Restricting now to rational transfer functions, we have the following result.

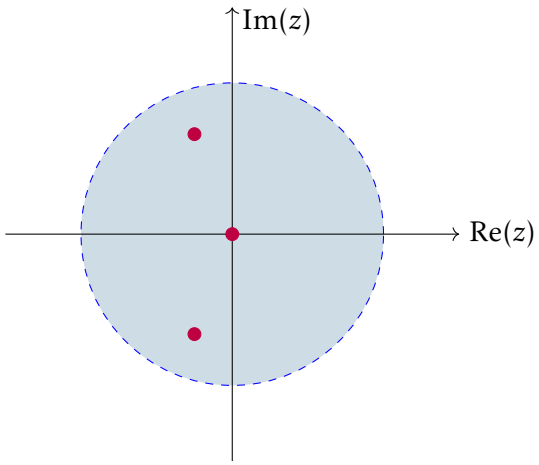


Figure 9: The poles of the transfer function $G(z)$ in eq. (5.80).

Theorem 5.3. A discrete causal LTI system with rational transfer function $G(z)$ is stable if and only if the poles of $G(z)$ belong to the set $\mathbb{D} = \{z : |z| < 1\}$.

Example 5.6. Consider the LTI system governed by the difference equation

$$2y_{n+3} + y_{n+2} + y_{n+1} = x_n, \tag{5.79}$$

subject to $y_0 = y_1 = y_2 = 0$. Transforming this equation leads to

$$Y(z)(2z^3 + z^2 + z) = X(z) \Longrightarrow G(z) = \frac{1}{z(2z^2 + z + 1)}. \tag{5.80}$$

Here we've identified the transfer function $G(z)$ by setting $X(z) = 1$. The poles of $G(z)$ occur when $z = 0$ and $2z^2 + z + 1 = 0$. Using the quadratic formula on the latter gives

$$z_\pm = \frac{-1 \pm \sqrt{1 - 4 \cdot 2 \cdot 1}}{4} = -\frac{1}{4} \pm \mathrm{i}\frac{\sqrt{7}}{4}. \tag{5.81}$$

In order to check the stability of the system, we need to see if the poles have modulus less than 1. The pole at $z = 0$ is clearly fine, while the modulus of $z_\pm$ is given by

$$|z_\pm| = \sqrt{\frac{1}{16} + \frac{7}{16}} = \frac{1}{\sqrt{2}} < 1. \tag{5.82}$$

Therefore, the system is stable.